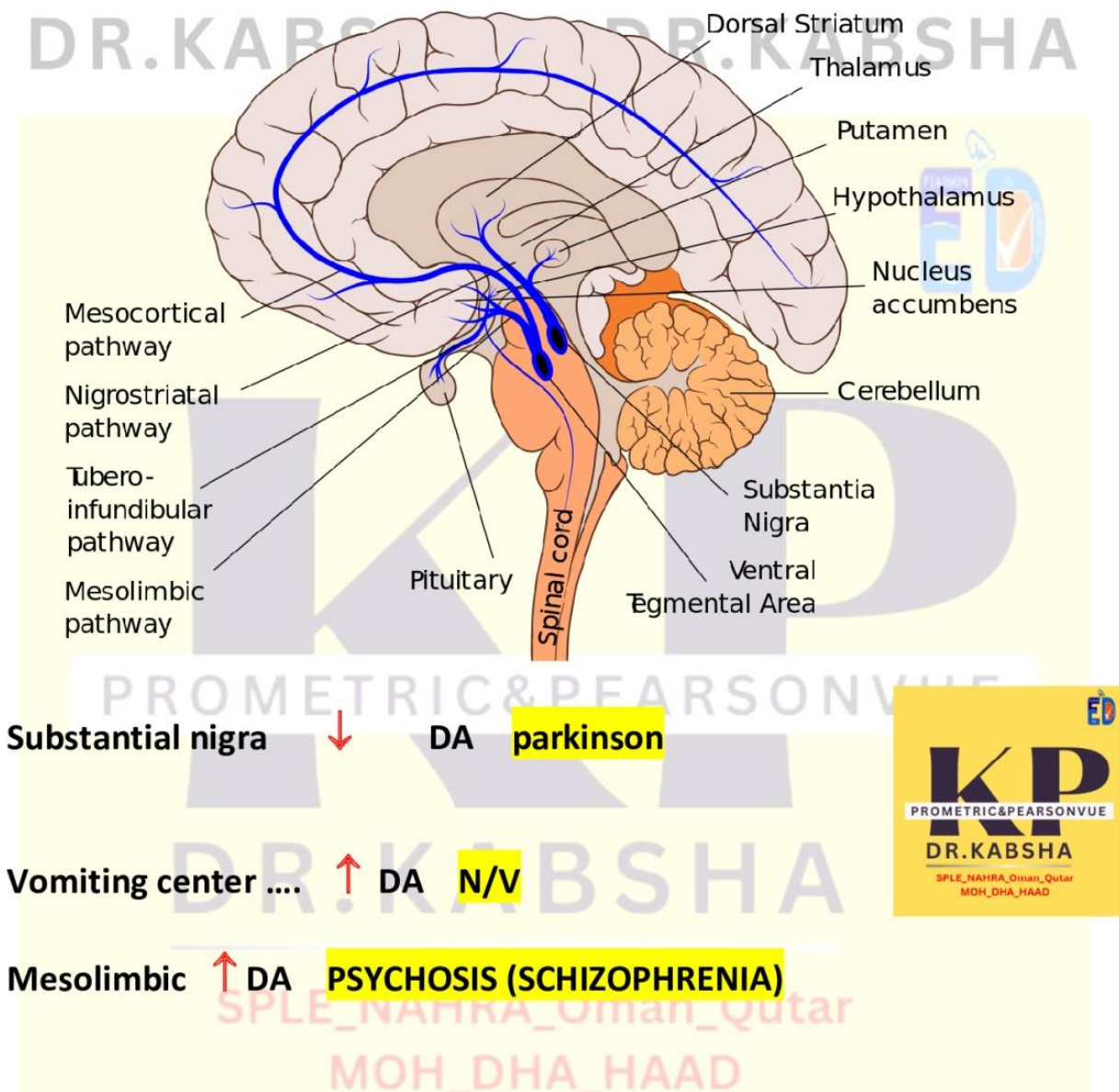




CSF FORMED IN? MOST CSF IS FORMED IN THE CEREBRAL VENTRICLES.
POSSIBLE SITES OF ORIGIN INCLUDE THE CHOROID PLEXUS,

Myelin is an insulating layer?? Protection & conduction

ANTIPSYCHOTICS

Typical Antipsychotics (1st generation) –

+ve symptoms

DA 

Extrapyramidal side effect (EPS)

A.Low potency

Chlorpromazine... pigment retina
Thioridazine

B.High potency

Haloperidol,,pimozide,promethazine,
Fluphenazine

QT prolongation?

Atypical Antipsychotics (2nd generation)

-ve symptoms

DA  **, 5-HT** 

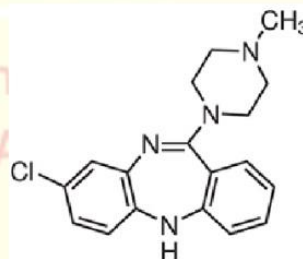
Metabolic side effect (wt,,D.M,,lipid)

Clozapine, Aripiprazole, olanzapine,

Quetiapine,ziprasudine , lurasidone

Respiridone?

Clozapine? Agranulocytosis



Monitor ANC, WBC like?

C.I seizures

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- Medication for Extrapyrimal effect: **Benztrapine, Diphenhydramine, BZDs**
- Pt takes olanzapine with high glucose level?
- Pt with involuntary movement in face. Cause?
- Antipsychotic for Parkinson patient?
- Smoking effect on olanzapine?

Increase metabolism decrease med. Conc. Increase CL of Med.

- Pt with Tardive Dyskinesia (TD):

Valbenazin # 1st FDA medication approved for TD

Parkinson's disease (PD)#

❖ **Treatment**

A. Levo dopa/Carbidopa:

Dopa precursor of dopamine

MOA of L-dopa: DA in brain, leading to stimulate DA receptor

L-dopa -COOH → dopamine × BBB

Dopa decarboxylase enzyme in brain & peripheral

Carbidopa <<< peripheral dopa decarboxylase

Take it on **empty** (morning)?? Like?

B. Dopamine receptor agonist:

Pramipexole, Ropinirole, Apomorphine, Rotigotine, Bromocriptine (used in Hyperprolactinemia)

CI: Do not use with 5HT3 antagonists (ondansetron, others) due to severe hypotension and loss of consciousness

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C.Catechol-O-Methyltransferase (COMT) inhibitors:

Tolcapone# Hepatotoxic

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D.Amantadine:

Antiviral, Antiparkinsonian, DA agonist # cause Livedoreticularis
"pigmentation"

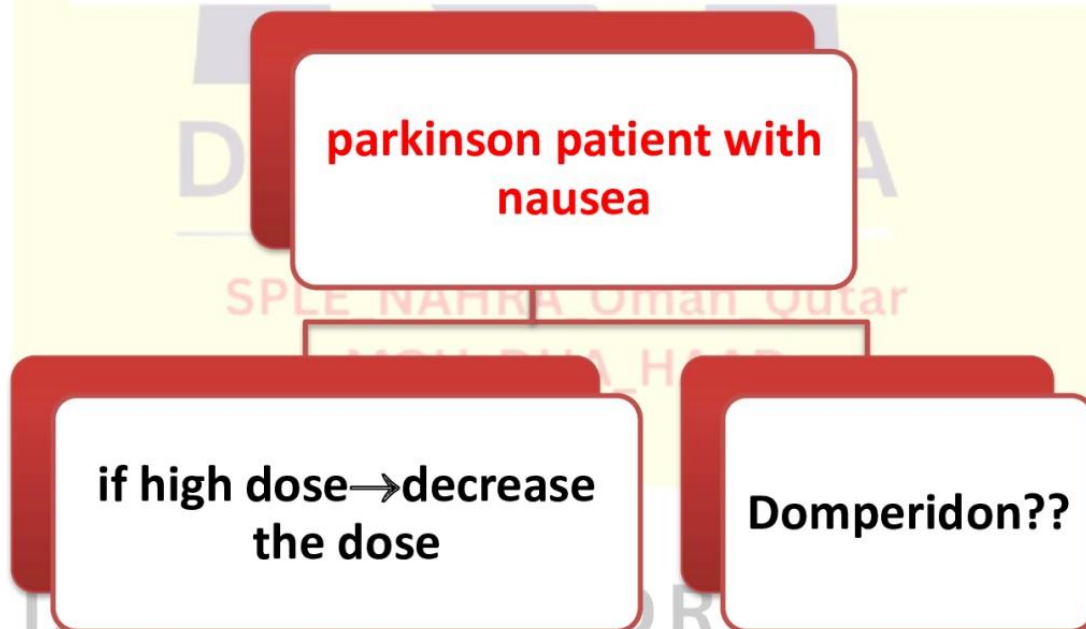
MOA of Amantadine: interfere with viral M2 protein function,
blocking uncoating of the virus particles هام

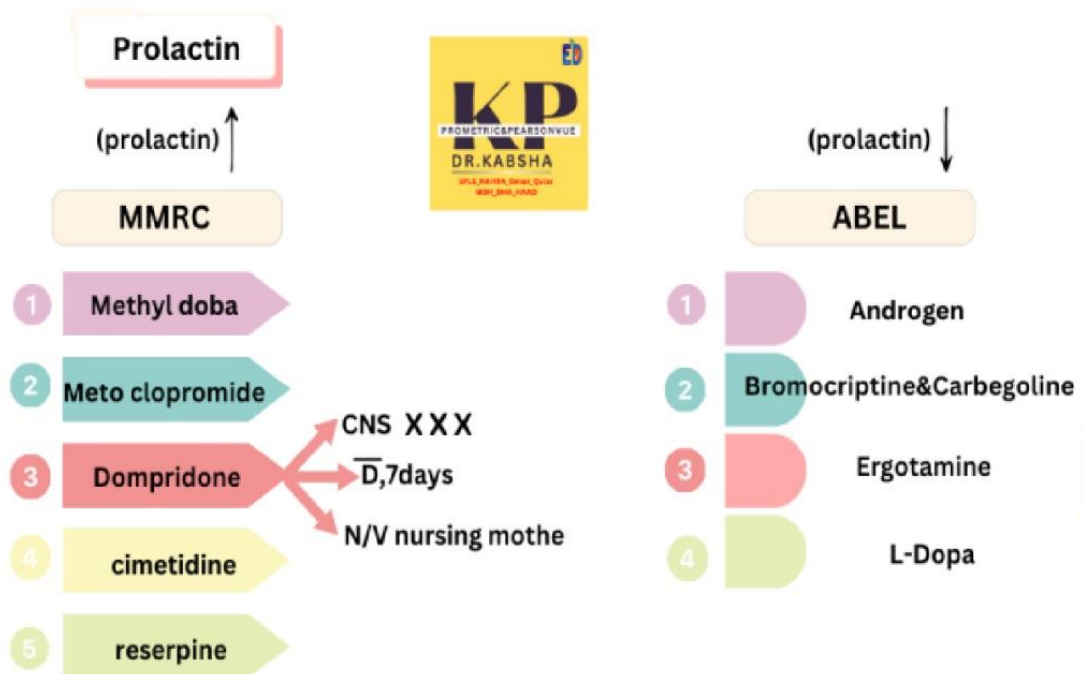
E. Anticholinergic:

Trihexyphenidyl,,benzotropine ,, biperidine

F . MAO-B inhibitor :

selegline , rasagiline





❖ **Classification of antidepressants :**

- MAOI
- TCA
- SSRI
- Atypical

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MOAI: inhibit the enzyme monoamine oxidase, which breaks down catecholamines, including 5-HT, NE, EPI, and DA. If these NTs increase dramatically, hypertensive crisis, and death can result

notes.

- MAO **A** in brain & peripheral (liver) non specifically on NA, 5-HT, DA
- MAO **B** CNS only act on DA only



Irreversible

reversible

Phenelzine

moclobemide

Tranylcypromine

Isocarboxazid

S/E

- **drug food interaction (cheese reaction) HTN crisis,**
- **serotonin syndrome**

Cheese contain (tyramine) catecholamine releasing agent

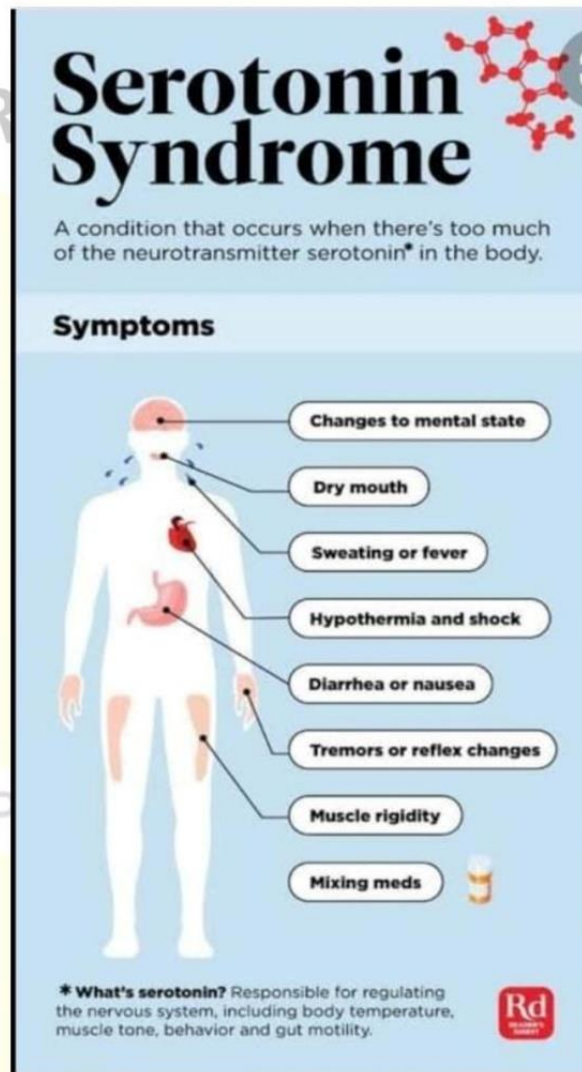
& MAOI inhibit metabolism of catecholamine

Aviod :

3. Selective MAO B inhibitor (selegiline & rasagiline)

* wash out 2 weeks

*Wash out 5-6 weeks in case of fluoxetine (long acting)



(Serotonin syndrome) antidote: cyproheptadine, BDZ

Note **linezolid** antibiotic with MAOI ?????? Exam

Tablet Selegline for Parkinson disease (PD), Transdermal patch for depression

Tricyclic antidepressants (TCA)

Secondary amines → Nortriptyline, Desipramine

Tertiary amines → Amitriptyline, Doxepin, Clomipramine, Imipramine

Tetracyclic antidepressant: maprotiline, amoxapine

MOA: primarily inhibit NE and 5-HT reuptake. They also block ACh and histamine receptors.

SE: Cardiotoxicity, Anticholinergic: Dry mouth, blurred vision, urinary retention, constipation, weight. Quinine like effect

CI: Concurrent use with MAO inhibitors (**serotonin syndrome**)

❖ **SSRI**

- Citalopram ...40mg ↑ QT
- Fluvoxamine ... OCD ttt.
- Fluoxetine Longer t_{1/2} (need 4 weeks wash OUT time) ..bulimia nervosa ttt.
- Sertraline safe myocardial infarction

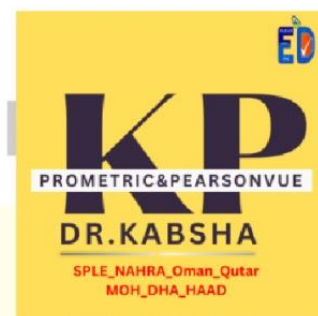
S/E... loss of libido & delayed ejaculation الحل؟؟

- ↓ Dose
- Bupropion (DOC)
- Sildenafil..cyproheptadine H& 5-HT

■ **SNRI**

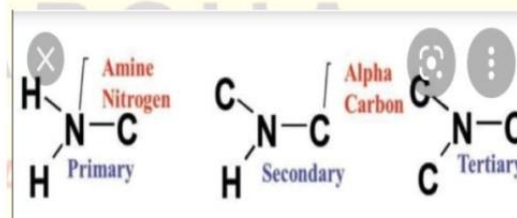
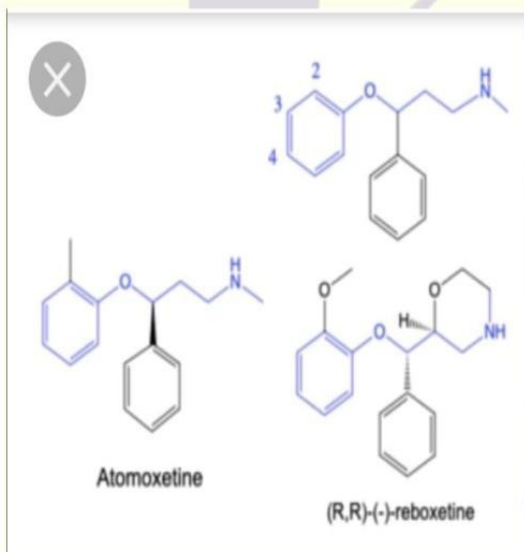
→ Venlafaxine...

→ Duloxetinettt .diabetic neuropathy

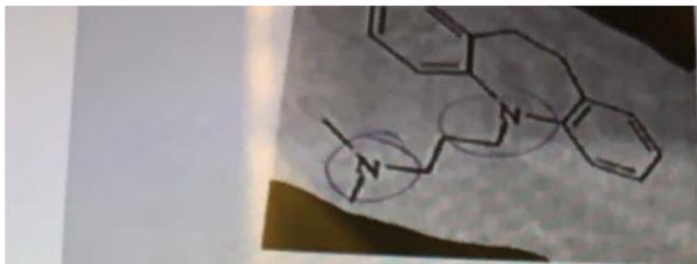


■ **NRI**

Reboxetine exam Q structure??



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49- The structure of a norepinephrine-serotonin reuptake inhibitor (NSRI) (see image).

Which of the following structural modifications can transform this drug into a Selective Norepinephrine Re-uptake Inhibitor (SNRI) analogue?

- A. Quaternization of the side chain amino group
- B. Transformation of the side chain amino group into a primary amine
- C. Replacement of the side chain N, N-dimethylamino group with a hydroxyl group
- D. Demethylation of the side chain tertiary amino group to yield a secondary amino group

SPLE_NAHRA_Oman_Qutar
MOH_DHA_HAAD

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Atypical anti-depressant:

Bupropion NE, DA reuptake

❖ use

- ✚ Depression + obese **but** Depression + thin ...mirtazapine
- ✚ Smoking cessationalso varenicline
- ✚ Sexual problem due to SSRI
- ✚ Not used in **epilepsy** **but**

○ Depression + epilepsy (psychosis), anxiety → paroxetine (SSRI)

○ Depression + pregnant, nursing mother → sertraline (SSRI)

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▪ **Alzheimer disease (AD)**

Treatment

This damage initially takes place in parts of the brain involved in memory, including the **entorhinal cortex and hippocampus**. It later affects areas in the **cerebral cortex**

A. Cholinesterase inhibitors: Donepezil, Rivastigmine, Galantamine

Inhibit acetyl cholinesterase (the enzyme responsible for hydrolysis of Ach). This **result increase Ach**

- ★ **Mild - Moderate:** Donepezil 5-10mg, Rivastigmine, Galantamine
- ★ **Advanced:** **ONLY** Donepezil 23 mg bed time
- ★ **Dementia+ Alzheimer disease (AD):** **ONLY** Rivastigmine

B. N-methyl-d-aspartate receptor antagonist: Memantine

✚ **MOA of Memantine:** antagonist effect at 5HT3 receptor, **NMDA receptor** & block nicotine receptor

- ✚ **All stages:** Donepezil
- ✚ **Moderate-sever:** Memantine +/- Donepezil
- ✚ **Herbal for Alzheimer disease:** Gingko biloba, Caprylidene, vit E, vit D
- ✚ **Drugs worsen dementia :** BBT, anticholinergic, antipsychotics

A 79 year-old man with Alzheimer's disease has mild to moderate dementia and is on oral rivastigmine 6 mg twice daily. He is not able to tolerate the medication due to nausea and vomiting. What would be the best recommendation?

- A. Suppositories of prochlorperazine 25 mg 60 minutes before each dose of Rivastigmine
- B. Discontinue Rivastigmine and substitute with Galantamine 4 mg twice
- C. daily Metoclopramide 10 mg 30 minutes before each dose of
- D. Rivastigmine Change to transdermal application of Rivastigmine 4.6 mg/24 hours

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Lithium: mood stabilizer

S/E :

Tremors → propranolol

Convulsion → Phenobarbital ,BDZ

Hypothyroidism → levothyroxine

Renal problem

monitoring 6-12 months

Li antidote : NaHco₃, Na polystyrene

Li toxicity due to hyponatremia

C.I pregnancy 1st trimester.. Breastfeeding??

Monitoring: 30 min before dose, drawn 8-12 hours post dose,

Take it with food to decrease nausea like?

Acute: A. Manic: Valproate or Lithium + Antipsychotic
B. Bipolar: Lithium or lamotrigine

Maintenance: A. Manic: Lithium +/- 2nd generation of Antipsychotic
B. Bipolar: Lamotrigine

BDZ

antidote: flumazenil

Potentiate action of GABA by binding to BDZ site on GABA receptor
..Increase affinity their affinity for GABA. ↑ Frequency of opening of
cl channel..Sedative effect

Long acting: Diazepam, clonazepam, clorazepate

Muscle spasm seizure emergency iv

Intermediate: lorazepam , chlordiazepoxide

Alcohol withdrawal symptoms

Short acting: alprazolam, midazolam, triazolam

S/E ..rebound insomnia ,respiratory depression anterograde amnesia

- **Drowsiness& confusion**common with **long** acting
- **Tolerance & dependence**common with **short** acting

Z drug

Low risk of respiratory depression & Tolerance & dependence

Zaleplon ...short term ttt insomnia

Zolipdem & eszopiclone ...chronic insomnia

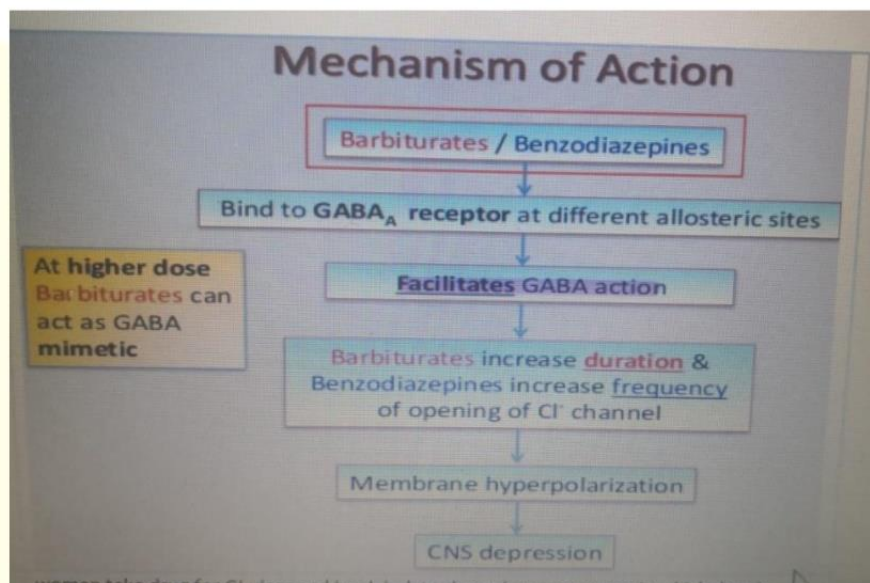
Zolipdem dose

Man 3.5 mg

max 10mg Woman 1.75 mg

max 5mg

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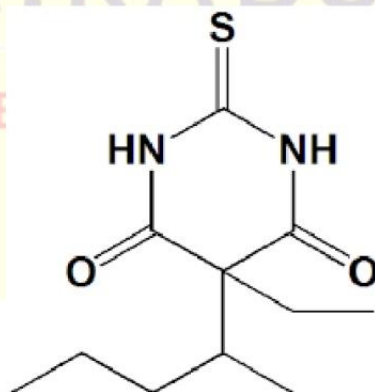


Long acting: Phenobarbital..primidone

Intermediate: secobarbital ,amobarbital

Ultra short: thiopental

Note: Primidone metabolize to Phenobarbital & phenylethylmalonamide (all active)



Thiopental

Epilepsy

• Seizure types

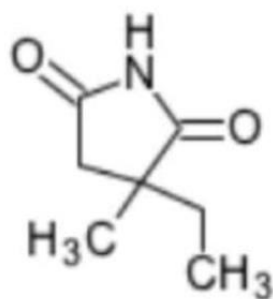
1-Focal seizures

Start on one side of the brain

2-Generalized seizures

- **Tonic clonic**
- **Absence (petit mal) (children)**
- **Myoclonic**
- **Atonic**
- **Toinc**
- **cloinc**

Ethosuximide only in absence by blocking t type ca channel blocker



D00539



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Status epilepticus (SE)

Any seizure that lasts more than 5 mins or more than 1 seizure within 5 min without returning to normal between them.

Ttt

1st iv lorazepam, IM midazolam, rectal diazepam
2nd fosphenytoin or Levetiracetam



Medication, ,,,, target ? Diet?

➡ Sodium Channel Blockers (Na⁺): -

- Phenytoin, Fosphenytoine : HLA-B 1502??
- Carbamazepine, Oxecarbamazapin(hyponatremia)
Carbamazepine

1st line in focal & generalized tonic clonic ,Used in trigeminal neuralgia,

- Lamotrigine (skin rash) metabolized by glucuronic conjugation

➡ GABA enhancing agent (GABA): -

- Barbiturate. # withdrawal symptoms - Benzodiazepine (BZDs) –
- Valproic acid (alopecia) antidote ?

➡ GABA analog

- Gabpentine
- Pregabalin

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• **Phenytoin**

Enzyme inducer,,, HLA-B 1502,,

Strong correlation between the plasma level and its effect

Dose:

phenytoin sodium **100 mg = 92 mg** of phenytoin base

Phenyntion & ibuprofen

S/E

- **Dose related :** nystagmus ,ataxia
- **Non dose related :** gingival hyperplasia(but no with ethotoin),,hirsutism , purple glove syndrome , phelibitis(but no with fosphenytoin)
Hepatotoxicity
- **Pregnancy :** category D

Fetal hydantoin syndrome : clift lip , heart problem

Fetus bleeding : decrease vit k

PLASMA LEVEL (MICROGRAMS/ML)	SIDE EFFECTS
<10	Usually none
10-20	Occasional mild nystagmus
20-30	Nystagmus
30-40	Ataxia, slurred speech, nausea and vomiting
40-50	Lethargy, confusion
>50	Coma, seizures

$$\text{Corrected phenytoin} = \frac{\text{Phenytoin observed}}{(\text{Albumin} \times 0.2) + 0.1}$$

phenytoin infusion rate not exceed??

- A-30mg per minute
- B-40 mg per minute
- C-50mg per minute
- D-100mg per minute

**Maximum dose of
phenytoin?**

- A. 300mg/D
- B. 400mg/D
- C. 500mg/D
- D. 600mg/D

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Others:

- lacosamide PR interval
- vigabatrin Vision loss??
- zonisamide oilgo hydrosis ,, nephrolithiasis
- topiramate

The screenshot shows a medical application interface. At the top, there is a title bar with a close button (X), the title "Topiramate", and two circular icons (a camera and a menu). Below the title, it says "toe PYRE a mate". Underneath, there is a section titled "Mechanism of action" with a list of likely multiple mechanisms:

- Likely multiple, including:
 - a. Blockade of voltage-gated Na⁺ and Ca⁺⁺ channels
 - b. Potentiation of inhibitory effects of GABA at GABA-A receptors.
 - c. Blockade of AMPA glutamate receptors.
 - d. Inhibition of carbonic anhydrase.

Max dose 400mg /d

Crcl < 70 ml / min ... require dose adjustment

Metabolic acidosis ??? wt loss

Levetiracetam

Ca channel blocker + GABA Max 3000mg /d

Crcl <80 ...dose adjustment

Safe for ??

Note

Treat neuropathic pain and depression: 1st: duloxetine,
pregabalin Others: Gabapentin, venlafaxine, CBZs,

Antiepileptic for pregnant Levetiracetam, lamotrigine

Antiepileptic for breastfeeding &: Levetiracetam A

antiepileptic need renal adjustment: Topiramate

Antiepileptic in absence seizures (petit): Ethosuximide

for older + (LOT): lorazepam, Oxazepam, temazepam?

- Pregnant woman take valproic acid with folic acid

ADHD

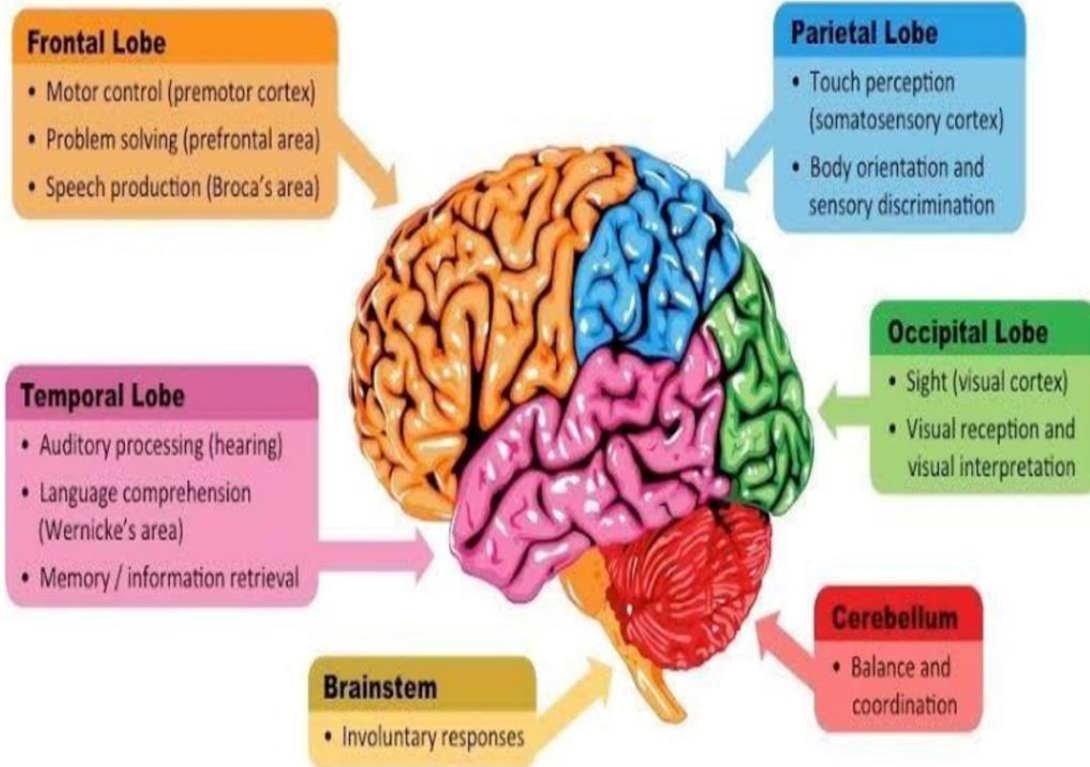
Methylphenidate extended release once daily

Amphetamine Cns stimulant.. Indirect sympathomimetic act indirectly to increase the concentration of the endogenous neurotransmitter by causing its release

Multiple Sclerosis (MS):: (oral) # Fingolimod

DC 2 months before getting pregnant & during pregnancy

Cardiotoxicity????



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MOH_DHA_HAAD

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